Weighted Adaptive Stimulation After Batch-testing Identification (WASABI) Problem Formulation

Problem Formulation

There are a few ways we can formulate the question

1. Similar to Draelos 2020 first identify the binary connections using group testing, then use the identified connections to fit a weighted model.
2. Directly use a linear-Poisson model to identify weights using group testing. (Works on the assumption that connections are linear)
3. If not linear, use a neural network or some other nonlinear approach.

We will focus on (2) first, with fixed continuous stimulation parameters (e.g., current, frequency...)

Goal

To develop a method for adaptive stimulation that iteratively identifies the most informative stimulation sites. The dynamics of the reach is modeled using a linear-Poisson model , using a weighted linear-Poisson model to identify the most informative stimulation sites and improve the accuracy of the model.

Dynamics model

We model stimulation as a perturbation to a latent dynamical system through the mapping $\boldsymbol{B}$. Specifically, we model the system using a Poisson linear dynamical system (PLDS), which combines firing rate estimation with dimensionality reduction. We assume linear, Markovian (AR(1)) state evolution, Poisson emissions, and stationary $\boldsymbol{A}$. Eventually, we can have a Kalman-like update to allow for nonstationarity in $\boldsymbol{C}$ and a compositional model for $\boldsymbol{A}$.

State space equation

$$
\begin{aligned}
&\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
&\boldsymbol{x}_t = \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{B}\,\mathrm{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
&\boldsymbol{y}_t \sim \mathrm{Poisson}\big(\mathrm{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)
\end{aligned}
$$

Variables

- $\boldsymbol{x}_t \in \mathbb{R}^K$: latent state at time t
  - K is the latent dimensionality
- $\boldsymbol{y}_t \in \mathbb{Z}_{\geq 0}^N$: binned spike count vector at time t
  - N is the number of channels
- $\boldsymbol{s}_t \in \{0,1\}^N$: one-hot stim indicator at time t
  - zero vector on control trials
  - stimulation occurs at $t = t_{\text{stim}}$

Hyperparameters

- $K \in \mathbb{Z}_+$: latent dimensionality
  - $K \ll N$, low-rank assumption
- $w \in \mathbb{R}_+$: stim spatial spread radius
- $\boldsymbol{V} \in \mathbb{R}^{N \times N}$: Toeplitz matrix encoding spatial current spread

Fitted parameters

- $\boldsymbol{A} \in \mathbb{R}^{K \times K}$: dynamics operator
- $\boldsymbol{B} \in \mathbb{R}^{K \times N}$: stimulus-to-latent mapping
- $\boldsymbol{b} \in \mathbb{R}^K$: perturbation threshold
  - what stim strength would perturb latents
- $\boldsymbol{C} \in \mathbb{R}^{N \times K}$: observation (loading) matrix
- $\boldsymbol{d} \in \mathbb{R}^N$: per-channel mean log-firing rate
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\boldsymbol{\mu}_0 \in \mathbb{R}^K$, $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{K \times K}$: initial state mean and covariance
- $\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{b}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\}$: full parameter set

$$
V_{ij} = \frac{1}{Z(w)} \exp\left(-\frac{(i-j)^2}{2w^2}\right) \quad Z(w) = \sum_{k=-\infty}^{\infty} \exp\left(-\frac{k^2}{2w^2}\right)
$$

Overview

1. **Control trials**: Monkey performs control reaches to the target
2. We fit the PLDS to the control reaches to obtain $\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}$
3. **Stim trials**: During each trial, perform online filtering using the fitted parameters $\hat{\boldsymbol{\theta}}_{\text{i}-1}$, to get $\{\boldsymbol{x}_{i,1:t}\}$ in real time. This gives us the state at which the stimulation happens.
   (a) At the first trial, we initialize with $\hat{\boldsymbol{\theta}}_{\text{i}-1} = \{\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{B}}_{0|0}, \hat{\boldsymbol{b}}_{0|0}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}, \hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\Sigma}}_0\}$, where $\hat{\boldsymbol{B}}_{0|0} = \boldsymbol{0}$, $\hat{\boldsymbol{\Sigma}}_{B_{0|0}} = \sigma_B^2 \boldsymbol{I}$, and $\hat{\boldsymbol{b}}_{0|0} = \boldsymbol{0}$.
   (b) This means that we have no information about the stim-to-latent mapping or the perturbation threshold and the first stimulation sites are chosen at random.
4. **Model fitting**: Between each trial, we fit the spike counts $\{\boldsymbol{y}_{i,1:T}\}$ to the model using the Extended Kalman Filter (EKF)-style integral and stochastic gradient descent (SGD). This gives us updated parameters $\hat{\boldsymbol{\theta}}_{\text{i}} \leftarrow \hat{\boldsymbol{\theta}}_{\text{i}-1}$
   (a) The previous trial would "warm-start" the convergence of the next trial, so we only need to run a few iterations of EKF and SGD between each trial to get a converging estimate of the parameters.
5. **Acquisition:** Choose the next stimulation site $\boldsymbol{s}_{i+1}$ using the covariance estimate of $\hat{\boldsymbol{B}}_{i|i}$ ($\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$), which is the uncertainty in the stim-to-latent mapping after trial i.
6. Perform the next stimulation and repeat 3-5.

Model fitting

State-space equation (PLDS)

$$
\begin{aligned}
&\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
&\boldsymbol{x}_t = \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{B}\,\mathrm{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
&\boldsymbol{y}_t \sim \mathrm{Poisson}\big(\mathrm{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)
\end{aligned}
$$

We denote two distinct indices throughout: Ex: for $\boldsymbol{A}_{t,i}$

- $i = 1, 2, \ldots$ indexes *trials*
- $t = 1, \ldots, T$ indexes *within-trial timesteps*

Initialization (priors at first stim trial)

We first collect a calibration session with the monkey performing control reaches, which identifies the initial model parameters (and spike detection templates).

Control trial initialization ($i = 0$)

Initialize and fit PLDS on control trials:

$$
\begin{aligned}
\hat{\boldsymbol{A}}_{\text{ctrl}} &= \boldsymbol{I}_K \\
\hat{\boldsymbol{C}}_{\text{ctrl}} &\sim \mathcal{N}(\boldsymbol{0},\, \sigma_C^2\, \boldsymbol{I}_{N\times K}) \\
\hat{\boldsymbol{d}}_{\text{ctrl}} &= \log \bar{\boldsymbol{y}}_{\text{ctrl}} \\
\hat{\boldsymbol{\mu}}_0 &= \boldsymbol{0}_K \\
\hat{\boldsymbol{\Sigma}}_0 &= \boldsymbol{I}_K
\end{aligned}
$$

$\Downarrow$ fit on control data

$$
\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}
$$

First stim trial initialization ($i = 1$)

Warm start with fitted control parameters:

$$
\begin{aligned}
\hat{\boldsymbol{A}}_1 &= \hat{\boldsymbol{A}}_{\text{ctrl}} \\
\hat{\boldsymbol{C}}_1 &= \hat{\boldsymbol{C}}_{\text{ctrl}} \\
\hat{\boldsymbol{d}}_1 &= \hat{\boldsymbol{d}}_{\text{ctrl}}
\end{aligned}
$$

$\hat{\boldsymbol{B}}_{0|0}$ and $\hat{\boldsymbol{\Sigma}}_{B_{0|0}}$ initialized via:

$$
\begin{aligned}
\hat{\boldsymbol{B}}_{0|0} &= \boldsymbol{0}_{K\times N} \\
\hat{\boldsymbol{\Sigma}}_{B_{0|0}} &= \sigma_B^2\, \boldsymbol{I} \\
\hat{\boldsymbol{b}}_{0|0} &= \boldsymbol{0}_K
\end{aligned}
$$

where:

- σ_B^2: controls prior uncertainty over the stim-to-latent mapping
- σ_C^2: controls prior uncertainty over the latent-to-observation mapping
- $\bar{\boldsymbol{y}}_{\text{ctrl}}$: mean spike count from control trials
- $\hat{\boldsymbol{\Sigma}}_{B_{0|0}} = \sigma_B^2\, \boldsymbol{I}$ (spherical prior) means $\boldsymbol{s}_1$ is chosen at random

Stim trial initialization ($i \geq 2$) uses the previous trial's fitted parameters $\hat{\boldsymbol{\theta}}_{i-1}$ as the prior for the next trial.

Online forward filtering during a trial

During a trial, we estimate the latent state $\boldsymbol{x}_{i,t}$ in real time using the EKF forward pass with the current parameter estimates $\hat{\boldsymbol{\theta}}_{i-1}$. This gives us an estimate of the latent state in real time.

EKF forward pass

The marginal likelihood of the latent trajectory is:

$$\log p(\boldsymbol{y}_{1:T}|\boldsymbol{\theta}) = \log \int p(\boldsymbol{y}_{1:T} \mid \boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, p(\boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, d\boldsymbol{x}_{1:T}$$

$$\boldsymbol{y}_t \sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)$$

$$\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}\}$$

Because the Poisson emission $p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}; \boldsymbol{\theta})$ is non-Gaussian, the integral is intractable. At time $t-1$, we need to approximate it by linearizing the emission around $\hat{\boldsymbol{x}}_{t|t-1}$, the current filtered mean (latent state), yielding a Gaussian approximation to the predictive distribution

$$p(\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}; \boldsymbol{\theta}) \approx \mathcal{N}(\boldsymbol{\lambda}_{t|t-1},\, \boldsymbol{S}_t), \qquad \boldsymbol{\lambda}_{t|t-1} = \text{softplus}\Big(\boldsymbol{C}\hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{d}\Big)$$

where $\boldsymbol{S}_t$ is the estimated covariance.

$$\mathcal{L}(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_t|\, \boldsymbol{\lambda}_{t|t-1},\, \boldsymbol{S}_t)$$

Emission Jacobian

In the EKF formulation, we approximate the observation operator $\boldsymbol{H}_t$ with the Emission Jacobian described here: For $\boldsymbol{\lambda}_t = \text{softplus}(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d})$, we have the Jacobian (log-rate with respect to the latent state) is:

$$\boldsymbol{H}_t = \left.\frac{\partial \boldsymbol{\lambda}_t}{\partial \boldsymbol{x}_t}\right|_{\hat{\boldsymbol{x}}_t} = \text{diag}(\sigma(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d}))\, \boldsymbol{C} \in \mathbb{R}^{N \times K}$$

where $\sigma(\cdot)$ is the sigmoid function (derivative of softplus). We calculate the Emission Jacobian $\boldsymbol{H}_t$ to linearize the non-Gaussian Poisson emission and plug into the Kalman equations. This propagates the Gaussian estimations of latent states $\boldsymbol{x}_t$ one time step forward each, which we can plug in the Kalman equations.

EKF forward pass prediction update

$$\begin{aligned}
\hat{\boldsymbol{x}}_{t|t-1} &= \boldsymbol{A}\hat{\boldsymbol{x}}_{t-1|t-1} + \boldsymbol{B}\,\text{softplus}(\boldsymbol{V}\boldsymbol{s}_t + \boldsymbol{b}) \\
\boldsymbol{P}_{t|t-1} &= \boldsymbol{A}\boldsymbol{P}_{t-1|t-1}\boldsymbol{A}^\top + \boldsymbol{Q} \\
\boldsymbol{S}_t &= \boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top + \text{diag}(\boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{K}_t &= \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1} \\
\hat{\boldsymbol{x}}_{t|t} &= \hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{P}_{t|t} &= (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H}_t)\boldsymbol{P}_{t|t-1}
\end{aligned}$$

- $\hat{\boldsymbol{x}}_{t|t-1} \in \mathbb{R}^K$: predicted latent state mean
- $\hat{\boldsymbol{x}}_{t|t} \in \mathbb{R}^K$: filtered latent state mean
- $\boldsymbol{P}_{t|t-1} \in \mathbb{R}^{K \times K}$: predicted latent state covariance
- $\boldsymbol{P}_{t|t} \in \mathbb{R}^{K \times K}$: filtered latent state covariance
- $\boldsymbol{S}_t \in \mathbb{R}^{N \times N}$: innovation covariance
- $\boldsymbol{K}_t \in \mathbb{R}^{K \times N}$: Kalman gain
- $(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1})$: innovation (residual)
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\text{diag}(\boldsymbol{\lambda}_{t|t-1})$: Poisson observation noise
- $\boldsymbol{H}_t \in \mathbb{R}^{N \times K}$: locally linear approximation of emission matrix

Offline parameter update (Outline)

At the end of each trial i, the following steps are performed in order:

1. "E step" **EKF forward pass + RTS smoother**: Run the EKF forward over $\boldsymbol{y}_{i,1:T}$ with $\hat{\boldsymbol{\theta}}_{i-1}$ fixed, then run the RTS smoother backward to get filtered means and covariances $\{\hat{\boldsymbol{x}}_{i,1:T}, \boldsymbol{P}_{i,1:T}\}$. $\{\hat{\boldsymbol{x}}_{i,t|t}, \boldsymbol{P}_{i,t|t}\}_{t=1}^{T}$ is the lag-one cross-covariance needed for the M-steps below.

2. "M step" **Factorized estimation of parameters**: We factorize the estimation of parameters to get point estimates of parameters with no closed form ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$, $\hat{\boldsymbol{b}}_i$), but still getting covaraince estimates for those that we can using EM. This also keeps the faster convergence rate of the EM algorithm.

   (a) **Closed-form update $\hat{\boldsymbol{A}}_i$, $\hat{\boldsymbol{Q}}_i$**

   (b) **SGD update $\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$, $\hat{\boldsymbol{b}}_i$** Take one gradient step on the single-trial EKF marginal log-likelihood $\mathcal{L}_i = \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_{i,t} \mid \boldsymbol{\lambda}_{i,t|t-1}, \boldsymbol{S}_{i,t})$:

$$(\hat{\boldsymbol{C}}_i, \hat{\boldsymbol{d}}_i) \leftarrow (\hat{\boldsymbol{C}}_{i-1}, \hat{\boldsymbol{d}}_{i-1}) - \eta_i \nabla_{\boldsymbol{C},\boldsymbol{d}} \left[-\mathcal{L}_i(\hat{\boldsymbol{\theta}}_{i-1})\right]$$

   (c) **Update $\hat{\boldsymbol{B}}_i$** and $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$: Using the updated $\hat{\boldsymbol{A}}_i$, $\hat{\boldsymbol{Q}}_i$, $\hat{\boldsymbol{C}}_i$, and $\hat{\boldsymbol{d}}_i$ from steps 2-3, we update $\hat{\boldsymbol{B}}_i$ and get $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$

3. **Repeat 1 and 2 until convergence**:

   (a) Steps (a)-(c) are handled in the DYNAMAX package using conditional moments gaussian smoother (CMGS), but we lay out the details below.

   (b) We may need hyperparameter optimization on the batch size and learning rate.

   (c) This process can terminate early if the next trial happens before convergence.

   (d) We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation)

EKF forward pass + RTS smoother

The EKF forward pass is identical to the online version. (see **Online forward filtering**)

RTS smoother backward pass

$$\begin{aligned}
\boldsymbol{G}_t &= \boldsymbol{P}_{t|t}\boldsymbol{A}^\top \boldsymbol{P}_{t+1|t}^{-1} \\
\hat{\boldsymbol{x}}_{t|T} &= \hat{\boldsymbol{x}}_{t|t} + \boldsymbol{G}_t\Big(\hat{\boldsymbol{x}}_{t+1|T} - \hat{\boldsymbol{x}}_{t+1|t}\Big) \\
\boldsymbol{P}_{t|T} &= \boldsymbol{P}_{t|t} + \boldsymbol{G}_t\Big(\boldsymbol{P}_{t+1|T} - \boldsymbol{P}_{t+1|t}\Big)\,\boldsymbol{G}_t^\top
\end{aligned}$$

initialized at $\hat{\boldsymbol{x}}_{T|T}$, $\boldsymbol{P}_{T|T}$ and run backward for $t = T-1, \ldots, 1$.

EM closed-form update $\hat{\boldsymbol{A}}_i$, $\hat{\boldsymbol{Q}}_i$

Given the RTS-smoothed statistics $\{\hat{\boldsymbol{x}}_{i,t|T},\, \boldsymbol{P}_{i,t|T},\, \boldsymbol{P}_{i,t,t-1|T}\}_{t=1}^{T}$ from the E step, $\hat{\boldsymbol{A}}_i$ and $\hat{\boldsymbol{Q}}_i$ admit closed-form M-step updates identical to those of a standard linear-Gaussian LDS, since the state transition

$$\boldsymbol{x}_{i,t} = \boldsymbol{A}\boldsymbol{x}_{i,t-1} + \boldsymbol{B}\,\mathrm{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{i,t} + \boldsymbol{b}\big) + \boldsymbol{\epsilon}_t$$

is linear-Gaussian in $\boldsymbol{x}_{i,t}$ given the (currently fixed) stimulation input. Only the emission $\boldsymbol{y}_{i,t} \mid \boldsymbol{x}_{i,t}$ is non-Gaussian, so it does not enter this update.

Expected complete-data log-likelihood. The terms of the complete-data log-likelihood involving $\boldsymbol{A}$, $\boldsymbol{Q}$ are

$$\mathbb{E}\big[\log p(\boldsymbol{x}_{i,1:T})\big] \;\doteq\; -\frac{1}{2}\sum_{t=2}^{T}\mathbb{E}\Big[\big(\boldsymbol{x}_{i,t} - \boldsymbol{A}\boldsymbol{x}_{i,t-1}\big)^\top \boldsymbol{Q}^{-1}\big(\boldsymbol{x}_{i,t} - \boldsymbol{A}\boldsymbol{x}_{i,t-1}\big)\Big] - \frac{T-1}{2}\log\det\boldsymbol{Q}$$

where the expectation is taken under the smoothing posterior $p(\boldsymbol{x}_{i,1:T} \mid \boldsymbol{y}_{i,1:T}, \hat{\boldsymbol{\theta}}_{i-1})$, and $\doteq$ denotes equality up to additive constants independent of $\boldsymbol{A}, \boldsymbol{Q}$.

Sufficient statistics. Define the smoothed second-moment sums

$$\begin{aligned}
\boldsymbol{S}_{xx} &= \sum_{t=2}^{T}\Big(\boldsymbol{P}_{i,t-1|T} + \hat{\boldsymbol{x}}_{i,t-1|T}\hat{\boldsymbol{x}}_{i,t-1|T}^\top\Big) \\
\boldsymbol{S}_{x'x} &= \sum_{t=2}^{T}\Big(\boldsymbol{P}_{i,t,t-1|T} + \hat{\boldsymbol{x}}_{i,t|T}\hat{\boldsymbol{x}}_{i,t-1|T}^\top\Big) \\
\boldsymbol{S}_{x'x'} &= \sum_{t=2}^{T}\Big(\boldsymbol{P}_{i,t|T} + \hat{\boldsymbol{x}}_{i,t|T}\hat{\boldsymbol{x}}_{i,t|T}^\top\Big)
\end{aligned}$$

where $\boldsymbol{S}_{xx}$ uses the "earlier" timestep $t-1$ on both sides, $\boldsymbol{S}_{x'x}$ pairs the "later" timestep t with the earlier $t-1$, and $\boldsymbol{P}_{i,t,t-1|T}$ is the lag-one smoothed cross-covariance from the RTS backward pass above.

M-step. Maximizing the expected complete-data log-likelihood with respect to $\boldsymbol{A}$ gives a weighted least-squares solution; setting the gradient to zero yields

$$\hat{\boldsymbol{A}}_i \leftarrow \boldsymbol{S}_{x'x}\,\boldsymbol{S}_{xx}^{-1}$$

Substituting $\hat{\boldsymbol{A}}_i$ back in and maximizing with respect to $\boldsymbol{Q}$ gives the residual covariance

$$\hat{\boldsymbol{Q}}_i \leftarrow \frac{1}{T-1}\Big(\boldsymbol{S}_{x'x'} - \hat{\boldsymbol{A}}_i\,\boldsymbol{S}_{x'x}^\top\Big)$$

which is symmetrized in practice as $\frac{1}{2}(\hat{\boldsymbol{Q}}_i + \hat{\boldsymbol{Q}}_i^\top)$ to guard against numerical asymmetry.

Stim-trial caveat. If a stimulation pulse falls inside the window $[2, T]$, the transition $\boldsymbol{x}_{i,t_{\text{stim}}} = \boldsymbol{A}\boldsymbol{x}_{i,t_{\text{stim}}-1} + \boldsymbol{B}\,\boldsymbol{u}_{i,t_{\text{stim}}} + \boldsymbol{\epsilon}_t$ is no longer purely autonomous, so including $t = t_{\text{stim}}$ in $\boldsymbol{S}_{xx}, \boldsymbol{S}_{x'x}, \boldsymbol{S}_{x'x'}$ would bias $\hat{\boldsymbol{A}}_i$ toward absorbing the stim kick. The online, per-trial version of this update (see Online update loop) therefore excludes $t = t_{\text{stim}}$ from all three sums; the general form above is the version used when fitting on stim-free control data ($\hat{\boldsymbol{A}}_{\text{ctrl}}$, $\hat{\boldsymbol{Q}}_{\text{ctrl}}$).

SGD parameter update $\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$, $\hat{\boldsymbol{b}}_i$

Parameters related to the Poisson observation equation ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$) and the stim-to-latent threshold ($\hat{\boldsymbol{b}}_i$) are updated by differentiating through the EKF pass:

$$\boldsymbol{\theta}^{(\tau+1)} = \boldsymbol{\theta}^{(\tau)} - \eta_\tau \, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right]$$

where η_τ is the effective learning rate defined by the Adam optimizer. More specifically:

$$\begin{aligned}
\boldsymbol{m}_\tau &= \beta_1 \boldsymbol{m}_{\tau-1} + (1-\beta_1)\, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right] && \text{(momentum)} \\
\boldsymbol{v}_\tau &= \beta_2 \boldsymbol{v}_{\tau-1} + (1-\beta_2)\, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right]^2 && \text{(RMSprop)} \\
\hat{\boldsymbol{m}}_\tau &= \frac{\boldsymbol{m}_\tau}{1-\beta_1^\tau}, \qquad \hat{\boldsymbol{v}}_\tau = \frac{\boldsymbol{v}_\tau}{1-\beta_2^\tau} && \text{(bias correction)} \\
\boldsymbol{\theta}^{(\tau+1)} &= \boldsymbol{\theta}^{(\tau)} - \frac{\alpha}{\sqrt{\hat{\boldsymbol{v}}_\tau} + \epsilon}\, \hat{\boldsymbol{m}}_\tau && \text{(parameter update)}
\end{aligned}$$

where $\boldsymbol{m}_\tau$ and $\boldsymbol{v}_\tau$ are the first and second moment estimates, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ are the default hyperparameters, and α is the base learning rate. Gradients are computed via automatic differentiation in JAX in the DYNAMAX package.

EM closed-form update $\hat{\boldsymbol{B}}_i$ and $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$

Getting the posterior $p(\hat{\boldsymbol{B}}_{i|i-1} \mid \boldsymbol{y}_{1:i-1}, \boldsymbol{s}_i)$

Stimulation pattern $\boldsymbol{s}_i \in \{0,1\}^N$ is fixed for trial i and happens at t_{stim}.

$$\begin{aligned}
\boldsymbol{x}_{i,0} &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\boldsymbol{x}_{i,t} &= \boldsymbol{A}\boldsymbol{x}_{i,t-1} + \boldsymbol{B}\boldsymbol{s}_i + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
\boldsymbol{y}_{i,t} &\sim \text{Poisson}\Big(\text{softplus}\big(\boldsymbol{C}\boldsymbol{x}_{i,t} + \boldsymbol{d}\big)\Big)
\end{aligned}$$

We assume we can factorize across the $\boldsymbol{\theta}$s (Much like coordinate ascent variational inference). We get the estimates of $\boldsymbol{\theta} \setminus \boldsymbol{B}$ known from PLDS fit via EKF SGD from trials $1, \ldots, i-1$ and treated as fixed at trial i. $\boldsymbol{s}_t$ is known:

$$\boldsymbol{\theta} \setminus \boldsymbol{B} = \{\boldsymbol{A}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\} = \hat{\boldsymbol{\theta}}_{i-1} \setminus \boldsymbol{B}$$

$$\begin{aligned}
p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t}) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) \\
p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{B}, \boldsymbol{\theta} \setminus \boldsymbol{B}) \cdot p(\boldsymbol{B}) \cdot p(\boldsymbol{A}) \cdot p(\boldsymbol{C}) \cdots \quad \text{(by independence of } \boldsymbol{\theta} \text{ components)}
\end{aligned}$$

Absorbing $p(\boldsymbol{\theta} \setminus \boldsymbol{B})$ into the constant and expanding over $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) \propto \underbrace{p(\boldsymbol{B})}_{\text{Initialize via } \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_0^B)} \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{B}, \boldsymbol{\theta} \setminus \boldsymbol{B})}_{\text{Poisson emission}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{depends on } \boldsymbol{B}, \text{can get from EKF}} d\boldsymbol{x}_{1:t} \tag{1}$$

Likelihood after marginalizing out $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) = \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{\theta})}_{\text{Poisson emission, need EKF approx.}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{Gaussian dynamics prior}} d\boldsymbol{x}_{1:t}$$

$$\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\}$$

Bayes':

$$\underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t})}_{\text{posterior}} = \frac{\overbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\boldsymbol{y}_{1:t})}_{\text{evidence, constant in } \boldsymbol{\theta}}}$$

Acquisition function (Selecting next stimulation pattern)

We need to select the next stimulation pattern based on the updated posterior distribution of $\boldsymbol{B}$. Which means we need $\hat{\boldsymbol{B}}_{i+1|i}$ and $\hat{\boldsymbol{\Sigma}}_{B_{i+1|i}}$. We predict what the one timestep ahead prediction after stim for the next trial will be, to get $\hat{\boldsymbol{\Sigma}}_{B_{i+1|i}}$. Then we can choose stimulation $\boldsymbol{s}_{i+1}$ accordingly.

At time step $t-1$, one timestep before stimulation:

$$\begin{aligned}\boldsymbol{x}_{i+1,\mathrm{t}} &= \boldsymbol{A}\boldsymbol{x}_{i+1,t-1} + \boldsymbol{B}\boldsymbol{s}_{i+1} + \boldsymbol{\epsilon}_t \\ \boldsymbol{B}\boldsymbol{s}_{i+1} &= \boldsymbol{x}_{i+1,\mathrm{t}} - \boldsymbol{A}\boldsymbol{x}_{i+1,t-1} - \boldsymbol{\epsilon}_t\end{aligned}$$

Substitute $\hat{\boldsymbol{B}}_{i|i}$ as our estimator of $\boldsymbol{B}$, and substituting EKF smoothed means, poststim means $\hat{\boldsymbol{x}}_{i,t|T}, \hat{\boldsymbol{x}}_{i,t-1|T}$, in place of true latent states, and accurate description of estimations:

$$\hat{\boldsymbol{B}}_{i+1|i}\boldsymbol{s}_{i+1} = \hat{\boldsymbol{x}}_{i+1,t|T} - \hat{\boldsymbol{A}}\hat{\boldsymbol{x}}_{i+1,t-1|T} - \boldsymbol{\epsilon}_t$$

where:

- $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q})$
- $\boldsymbol{x}_{t-1} \approx \boldsymbol{x}_{t-1|T} \sim \mathcal{N}(\hat{\boldsymbol{x}}_{i,t-1|T}, \boldsymbol{P}_{i,t-1|T})$
- $\boldsymbol{x}_t \approx \boldsymbol{x}_{t|T} \sim \mathcal{N}(\hat{\boldsymbol{x}}_{i,t|T}, \boldsymbol{P}_{i,t|T})$
- For now we ignore $\mathrm{Cov}(\hat{\boldsymbol{A}})$

Now we want the covariance:

$$\begin{aligned}\boldsymbol{\Sigma}_{B_{i+1|i}} = \mathrm{Cov}(\hat{\boldsymbol{B}}_{i+1|i}) &= \mathrm{Cov}(\hat{\boldsymbol{x}}_{i,t|T}) + \mathrm{Cov}(\hat{\boldsymbol{A}}\hat{\boldsymbol{x}}_{i,t-1|T}) + \mathrm{Cov}(\boldsymbol{\epsilon}_t) \\ \boldsymbol{\Sigma}_{B_{i|i}} &\approx \mathrm{Cov}(\hat{\boldsymbol{A}}\hat{\boldsymbol{x}}_{i,t-1|T} + \hat{\boldsymbol{B}}_{i|i}\boldsymbol{s}_t + \boldsymbol{\epsilon}_t) + \hat{\boldsymbol{A}}\boldsymbol{P}_{i,t-1|T}\hat{\boldsymbol{A}}^\top + \boldsymbol{Q} \\ \boldsymbol{\Sigma}_{B_{i|i}} &\approx \mathrm{Cov}(\hat{\boldsymbol{B}}_{i|i}\boldsymbol{s}_t) + 2\hat{\boldsymbol{A}}\boldsymbol{P}_{i,t-1|T}\hat{\boldsymbol{A}}^\top + 2\boldsymbol{Q}\end{aligned}$$

Let $\hat{\boldsymbol{B}}_{i|i}\boldsymbol{s}_t$ be $\boldsymbol{B}\boldsymbol{s}$ for now.

$$\mathrm{Cov}(\boldsymbol{B}\boldsymbol{s}) = \mathbb{E}\Big[\boldsymbol{B}\boldsymbol{s}\boldsymbol{s}^T\boldsymbol{B}^T\Big] - \mathbb{E}[\boldsymbol{B}\boldsymbol{s}]\,\mathbb{E}[\boldsymbol{B}\boldsymbol{s}]^T$$

To capture the dependencies across channels (not IID), use the posterior of $\boldsymbol{B}$ from (1). We know from EKF $\mathbb{E}[\boldsymbol{B}] = \hat{\boldsymbol{B}}_{i|i}, \mathrm{Cov}(\hat{\boldsymbol{B}}_{i|i}) = \hat{\boldsymbol{\Sigma}}_{B_{i|i}}$:

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{B}\boldsymbol{s}) &= \mathbb{E}\Big[\boldsymbol{B}\boldsymbol{s}\boldsymbol{s}^T\boldsymbol{B}^T\Big] - \mathbb{E}[\boldsymbol{B}\boldsymbol{s}]\,\mathbb{E}[\boldsymbol{B}\boldsymbol{s}]^T \\ \mathrm{Cov}(\boldsymbol{B}\boldsymbol{s}) &= \mathbb{E}[\boldsymbol{B}]\,\boldsymbol{s}\boldsymbol{s}^T\mathbb{E}\Big[\boldsymbol{B}^T\Big] + \mathrm{Cov}(\boldsymbol{B})\cdot\boldsymbol{s}\boldsymbol{s}^T - \mathbb{E}[\boldsymbol{B}]\boldsymbol{s}\boldsymbol{s}^T\mathbb{E}[\boldsymbol{B}]^T \\ \mathrm{Cov}(\hat{\boldsymbol{B}}_{i|i}\,\boldsymbol{s}_t) &= \hat{\boldsymbol{\Sigma}}_{B_{i|i}}\,\boldsymbol{s}_t\boldsymbol{s}_t^\top\end{aligned}$$

Substituting back:

$$\hat{\boldsymbol{\Sigma}}_{B_{i|i}} \approx \hat{\boldsymbol{\Sigma}}_{B_{i|i}}\,\boldsymbol{s}_t\boldsymbol{s}_t^\top + 2\hat{\boldsymbol{A}}\boldsymbol{P}_{i,t-1|T}\hat{\boldsymbol{A}}^\top + 2\boldsymbol{Q}$$

Derivation

For adaptive selection of stimulation patterns, before each trial $i+1$, we want to select $\boldsymbol{s}_{i+1}$ that maximizes the expected information gain about $\boldsymbol{B}$. Equivalently, this would be selecting $\boldsymbol{s}_{i+1}$ that minimizes the determinant of the covariance of $\boldsymbol{B}$ after trial $i+1$ ($\hat{\boldsymbol{\Sigma}}_{B_{i+1|i+1}}$).

Let $\boldsymbol{\Sigma}^{-1}_{B_{i|i}}$	be the precision matrix of $\boldsymbol{B}$ after trial i
Let $\boldsymbol{\Delta}_{i+1} = \boldsymbol{B}\boldsymbol{s}_{i+1}$	be the latent perturbation induced by stimulation pattern $\boldsymbol{s}_{i+1}$
Let β	be how we scale our data relative to our prior (could be $1/n$)

In terms of precision, we can formulate the Gaussian case as:

$$\boldsymbol{\Sigma}^{-1}_{B_{i+1|i}} = \boldsymbol{\Sigma}^{-1}_{B_{i|i}} + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]^\top$$

Taking the determinant and applying the matrix determinant lemma:

$$\begin{aligned}
\det \boldsymbol{\Sigma}^{-1}_{B_{i+1|i}} &= \det\Big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}} + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]^\top\Big) \\
&= \Big(1 + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]\Big) \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}}\big)
\end{aligned}$$

Since we minimize over $\boldsymbol{s}_{i+1}$, the objective function is therefore:

$$\begin{aligned}
\arg\max_{\boldsymbol{s}}\ \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i+1|i}}\big) &= \arg\max_{\boldsymbol{s}}\ \Big(1 + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]\Big) \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}}\big) \\
&= \arg\max_{\boldsymbol{s}}\ \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}]
\end{aligned}$$

We can further expand out the expected perturbation in latent space:

$$\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{s}_{i+1}] = \hat{\boldsymbol{B}}_i \boldsymbol{s}_{i+1}$$

where $\hat{\boldsymbol{B}}_i$ is estimated from trials $1, \ldots, i$, giving the final objective:

$$\arg\max_{\boldsymbol{s}}\ \beta\, \boldsymbol{s}_{i+1}^\top \hat{\boldsymbol{B}}_i^\top \boldsymbol{\Sigma}^B_i\, \hat{\boldsymbol{B}}_i \boldsymbol{s}_{i+1}$$

TODO

1. We will need a **compositional model** of some sort to handle the fact that if there is behavioral change, there is likely a change in the latent dynamics as well, which will affect the estimation of $\boldsymbol{B}$. Change point detection? Or construct using known time-scale of behavioral / neural feedback
2. We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation):
3. We also want a **Kalman-like update** of the measurement equation parameters ($\boldsymbol{C}$ and $\boldsymbol{d}$)
4. Might need a **greedy search** for the acquisition function to not focus on the same noisy channels repeatedly.
5. This is for one timepoint, don't worry too much about some kind of t to t+k information update, before deciding on the next step, where t+k is the next stim LINOCS?
6. Need to expand to KxN for the precision matrix, instead of a K x K matrix. Current formulation gives you which latent to stim, which gives you probably channels just based on its contribution/composition of the latents
7. Tried fitting the PLDS with a step term ($\boldsymbol{C} \times \text{step}(t)$) before, but doesn't seem to do well.
8. Try to find closed form after plugging in $\mathbb{E}[\boldsymbol{B}]$?

- Since s is Bernoulli (discrete), projected or coordinate or proximal descent to solve for $\boldsymbol{s}$
- Projected is projecting the final to discrete, proximal is projecting each step

Random Sampling Stopgap

- Total configurations: N
- Take N samples: stimulate groups of 2 sites (one row).
- For 128 sites, this is 64 rows.
- Each stim randomly sample 20%: 12 rows (can revisit this number)
- Collect all the data and responses and see if we can predict in a leave-one-out approach (use all but one response).
- Benefit of random sampling is that we can collect now before Nike's implant comes out.

Online update loop (trial $i \geq 1$)

At the end of each trial i, the following steps are performed in order:

1. **EKF forward pass + RTS smoother.** Run the EKF forward over $\boldsymbol{y}_{i,1:T}$ with $\hat{\boldsymbol{\theta}}_{i-1}$ fixed, then run the RTS smoother backward:

$$\text{Filtered:} \quad \hat{\boldsymbol{x}}_{i,t|t} = \hat{\boldsymbol{x}}_{i,t|t-1} + \boldsymbol{K}_{i,t}(\boldsymbol{y}_{i,t} - \boldsymbol{\lambda}_{i,t|t-1}), \quad \boldsymbol{P}_{i,t|t} = (\boldsymbol{I} - \boldsymbol{K}_{i,t}\boldsymbol{H}_{i,t})\boldsymbol{P}_{i,t|t-1}$$

$$\text{Smoothed:} \quad \hat{\boldsymbol{x}}_{i,t|T} = \hat{\boldsymbol{x}}_{i,t|t} + \boldsymbol{G}_{i,t}(\hat{\boldsymbol{x}}_{i,t+1|T} - \hat{\boldsymbol{x}}_{i,t+1|t}), \quad \boldsymbol{P}_{i,t|T} = \boldsymbol{P}_{i,t|t} + \boldsymbol{G}_{i,t}(\boldsymbol{P}_{i,t+1|T} - \boldsymbol{P}_{i,t+1|t})\boldsymbol{G}_{i,t}^{\top}$$

   where $\boldsymbol{G}_{i,t} = \boldsymbol{P}_{i,t|t}\boldsymbol{A}^{\top}\boldsymbol{P}_{i,t+1|t}^{-1}$ is the smoother gain and $\boldsymbol{P}_{i,t,t-1|T} = \boldsymbol{P}_{i,t|t}\boldsymbol{G}_{i,t-1}^{\top}$ is the lag-one cross-covariance needed for the M-steps below.

2. **Closed-form update $\boldsymbol{A}$, $\boldsymbol{Q}$ (online EM M-step)** Exclude $t = t_{\text{stim}}$ to avoid absorbing the stim kick into $\boldsymbol{A}$:

$$\hat{\boldsymbol{A}}_i \leftarrow \left(\sum_{\substack{t=2 \\ t \neq t_{\text{stim}}}}^{T} \boldsymbol{P}_{i,t,t-1|T} + \hat{\boldsymbol{x}}_{i,t|T}\hat{\boldsymbol{x}}_{i,t-1|T}^{\top} \right) \left(\sum_{\substack{t=2 \\ t \neq t_{\text{stim}}}}^{T} \boldsymbol{P}_{i,t-1|T} + \hat{\boldsymbol{x}}_{i,t-1|T}\hat{\boldsymbol{x}}_{i,t-1|T}^{\top} \right)^{-1}$$

$$\hat{\boldsymbol{Q}}_i \leftarrow \frac{1}{T-2} \sum_{\substack{t=2 \\ t \neq t_{\text{stim}}}}^{T} \left[\boldsymbol{P}_{i,t|T} + \hat{\boldsymbol{x}}_{i,t|T}\hat{\boldsymbol{x}}_{i,t|T}^{\top} - \hat{\boldsymbol{A}}_i \left(\boldsymbol{P}_{i,t,t-1|T}^{\top} + \hat{\boldsymbol{x}}_{i,t-1|T}\hat{\boldsymbol{x}}_{i,t|T}^{\top} \right) \right]$$

3. **SGD update $\boldsymbol{C}$, $\boldsymbol{d}$.** Take one gradient step on the single-trial EKF marginal log-likelihood $\mathcal{L}_i = \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_{i,t} \mid \boldsymbol{\lambda}_{i,t|t-1}, \boldsymbol{S}_{i,t})$:

$$(\hat{\boldsymbol{C}}_i,\, \hat{\boldsymbol{d}}_i) \leftarrow (\hat{\boldsymbol{C}}_{i-1},\, \hat{\boldsymbol{d}}_{i-1}) - \eta_i \, \nabla_{\boldsymbol{C},\boldsymbol{d}} \left[-\mathcal{L}_i(\hat{\boldsymbol{\theta}}_{i-1}) \right]$$

   where $\eta_i \propto 1/i$ is a decaying learning rate.

4. **Update $\boldsymbol{B}$ (trial-level Kalman two-step).** Using the freshly updated $\hat{\boldsymbol{A}}_i$, $\hat{\boldsymbol{Q}}_i$, $\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$ from steps 2–3.

   Predict — propagate $\boldsymbol{B}$ uncertainty from trial $i-1$ to i, using the RTS-smoothed covariance $\boldsymbol{P}_{i-1,\,t_{\text{stim}}-1|T}$ from the previous trial (tighter than the filtered covariance since it uses all T observations):

$$\hat{\boldsymbol{\Sigma}}_{B_{i|i-1}} \leftarrow \hat{\boldsymbol{\Sigma}}_{B_{i-1|i-1}} \, \boldsymbol{s}_i \boldsymbol{s}_i^{\top} + 2\, \hat{\boldsymbol{A}}_i \, \boldsymbol{P}_{i-1,\,t_{\text{stim}}-1|T} \, \hat{\boldsymbol{A}}_i^{\top} + 2\, \hat{\boldsymbol{Q}}_i$$

   Update — condition on $\boldsymbol{y}_{i,\,t_{\text{stim}}}$ via EKF linearization using $\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$:

$$\boldsymbol{K}_i^{B} = \hat{\boldsymbol{\Sigma}}_{B_{i|i-1}} \, \boldsymbol{s}_i \, \boldsymbol{H}_{i,t_{\text{stim}}}^{\top} \left(\boldsymbol{H}_{i,t_{\text{stim}}} \, \hat{\boldsymbol{\Sigma}}_{B_{i|i-1}} \, \boldsymbol{s}_i \boldsymbol{s}_i^{\top} \boldsymbol{H}_{i,t_{\text{stim}}}^{\top} + \operatorname{diag}(\boldsymbol{\lambda}_{i,t_{\text{stim}}})^{-1} \right)^{-1}$$

$$\hat{\boldsymbol{B}}_{i|i} \leftarrow \hat{\boldsymbol{B}}_{i|i-1} + \boldsymbol{K}_i^{B} \left(\boldsymbol{y}_{i,\,t_{\text{stim}}} - \boldsymbol{\lambda}_{i,t_{\text{stim}}} \right) \boldsymbol{s}_i^{\top}$$

$$\hat{\boldsymbol{\Sigma}}_{B_{i|i}} \leftarrow \left(\boldsymbol{I} - \boldsymbol{K}_i^{B} \, \boldsymbol{H}_{i,t_{\text{stim}}} \, \boldsymbol{s}_i \boldsymbol{s}_i^{\top} \right) \hat{\boldsymbol{\Sigma}}_{B_{i|i-1}}$$

5. **Select next stimulation pattern.**

$$\boldsymbol{s}_{i+1}^{*} = \arg \max_{\boldsymbol{s} \in \{0,1\}^N,\, \|\boldsymbol{s}\|_0 = k} \; \boldsymbol{s}^{\top} \hat{\boldsymbol{B}}_{i|i}^{\top} \, \hat{\boldsymbol{\Sigma}}_{B_{i|i}} \, \hat{\boldsymbol{B}}_{i|i} \, \boldsymbol{s}$$

   Steps 2 and 3 are decoupled: $\boldsymbol{B}$ is updated via the Kalman mechanism and $\boldsymbol{\theta} \setminus \boldsymbol{B}$ via SGD, both conditioned on the same EKF pass from step 1. This introduces an approximation: the EKF in step 1 is run under $\hat{\boldsymbol{B}}_{i-1|i-1}$, so the filtered states used in step 2 reflect the previous trial's $\boldsymbol{B}$ estimate rather than the current one.

Posterior for $\boldsymbol{B}$ and predict covariance derivation

We treat $\boldsymbol{B}$ as a random variable and factorize across $\boldsymbol{\theta}$ components (analogous to CAVI), treating $\hat{\boldsymbol{\theta}}_{i-1} \setminus \boldsymbol{B}$ as fixed. The posterior after $i-1$ trials is approximated as Gaussian:

$$p(\boldsymbol{B} \mid \boldsymbol{y}_{1:i-1}, \boldsymbol{s}_{1:i-1}) \approx \mathcal{N}(\hat{\boldsymbol{B}}_{i-1|i-1},\, \hat{\boldsymbol{\Sigma}}_{B_{i-1|i-1}})$$

The Poisson emission is non-Gaussian so each update is approximated via EKF linearization at t_{stim}, maintaining the Gaussian form across trials.

Derivation of the predict covariance. Rearranging the dynamics at t_{stim} to isolate the $\boldsymbol{B}$ contribution:

$$\hat{\boldsymbol{B}}_{i|i-1}\, \boldsymbol{s}_i = \hat{\boldsymbol{x}}_{i,\, t_{\text{stim}}|t_{\text{stim}}-1} - \hat{\boldsymbol{A}}_i\, \hat{\boldsymbol{x}}_{i,\, t_{\text{stim}}-1|t_{\text{stim}}-1} - \boldsymbol{\epsilon}_{t_{\text{stim}}}$$

Taking the covariance, using $\mathrm{Cov}(\hat{\boldsymbol{B}}_{i-1|i-1}\boldsymbol{s}_i) = \hat{\boldsymbol{\Sigma}}_{B_{i-1|i-1}} \boldsymbol{s}_i \boldsymbol{s}_i^\top$ (derived below) and ignoring $\mathrm{Cov}(\hat{\boldsymbol{A}})$:

$$\hat{\boldsymbol{\Sigma}}_{B_{i|i-1}} \approx \hat{\boldsymbol{\Sigma}}_{B_{i-1|i-1}}\, \boldsymbol{s}_i \boldsymbol{s}_i^\top + 2\, \hat{\boldsymbol{A}}_i\, \boldsymbol{P}_{i-1,\, t_{\text{stim}}-1|T}\, \hat{\boldsymbol{A}}_i^\top + 2\, \hat{\boldsymbol{Q}}_i$$